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Prenatal Acetaminophen (Paracetamol) Use and the Risk of Autism and/or Attention-Deficit/Hyperactivity Disorder Among Sibling-Matched Cohorts

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IMPORTANCE Paracetamol (acetaminophen) is the first-line analgesic and antipyretic recommended globally during pregnancy. Observational studies have reported associations with increased risks of autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD) in offspring, raising public and clinical concern; however, these findings may be substantially confounded by unmeasured familial factors.

OBJECTIVES To assess the association between prenatal paracetamol exposure and risk of ASD and ADHD in offspring using a sibling-matched study design to control for familial confounding.

DESIGN, SETTING, AND PARTICIPANTS This population-based cohort study, conducted from January 1, 2001, to December 31, 2023, in Hong Kong, used deidentified electronic health records within a triangulation framework. From an initial cohort of 708 020 mother-child pairs (approximately 43.3% with prenatally paracetamol exposure), sibling-matched cohorts were constructed of children from families with discordant prenatal paracetamol exposure and sufficient follow-up (≥2 years for ASD and ≥5 years for ADHD). Data analysis was conducted from October 2, 2025, to March 24, 2026.

EXPOSURE Prenatal paracetamol exposure, identified from electronic dispensing records (British National Formulary [BNF], section 4.7.1) with data on drug name, strength, dosage, and prescription dates.

MAIN OUTCOMES AND MEASURES Incident diagnoses per the *International Classification of Diseases, Ninth Revision, Clinical Modification,* for ASD (code 299) or for ADHD (code 314) or a prescription for ADHD-specific medication licensed in Hong Kong (methylphenidate, atomoxetine, or lisdexamfetamine [BNF, section 4.4]).

RESULTS The final cohorts comprised 124 333 children for ASD analysis (mean [SD] age, 9.3 [4.7] years; 61 775 females [49.7%] and 62 558 males [50.3%]) and 97 285 for ADHD analysis (mean [SD] age, 7.6 [4.2] years; 48 455 females [49.8%], 48 830 [50.2%]). In the sibling-matched analyses, prenatal paracetamol exposure was not associated with the risk of ASD (adjusted hazard ratio [aHR], 1.00; 95% CI, 0.91-1.11) or ADHD (aHR, 1.01; 95% CI, 0.93-1.08). Null associations were consistent across exposure timing, cumulative dose, and usage patterns (sporadic, intermittent, persistent), and were robust in sensitivity analyses. However, positive associations were observed in conventional cohort analyses, and importantly, also in negative control analyses of prepregnancy exposure for ASD (HR, 1.12; 95% CI, 1.08-1.17) and ADHD (HR, 1.24; 95% CI, 1.20-1.28).

CONCLUSIONS AND RELEVANCE In this population-based cohort study, a sibling-matched analysis found no evidence of an association between prenatal paracetamol exposure and the risk of ASD or ADHD in offspring. The positive signals observed in conventional studies are likely attributable to residual familial confounding. These findings provide important reassurance regarding the safety of indicated paracetamol use during pregnancy.

Invited Commentary page 1111

Supplemental content

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Paracetamol (acetaminophen) is the most frequently administered medication for pain and fever during pregnancy and is recommended as a first-line treatment by international regulatory agencies. Professional obstetric societies advise that it remains a reasonable and appropriate choice.[1] However, as a substance that crosses the placental barrier, its potential impact on fetal neurodevelopment necessitates rigorous scientific evaluation.[2]

Recent public announcements, including a prominent statement from the US President, have intensified scrutiny of prenatal paracetamol use, with claims of an elevated risk of autism spectrum disorder (ASD) in offspring.[3] In response, regulatory bodies, including the Medicines and Healthcare Products Regulatory Agency (UK) and the World Health Organization, have reaffirmed its current safety profile for use during pregnancy.[4,5] This divergence between public concerns and established clinical guidelines underscores a critical gap, highlighting an urgent need for more definitive evidence.

A recent umbrella review[6] critically appraised the methodologic quality of existing systematic reviews on this topic, concluding that these syntheses are largely of low to critically low quality and exhibit a high degree of redundancy. Of the 6 primary studies on ASD,[7-12] 4 reported positive associations with prenatal paracetamol exposure.[7,8,10,12] However, the umbrella review[6] highlighted that these findings are inherently susceptible to residual confounding because essential familial and genetic factors were often insufficiently controlled for.[6] Furthermore, a key concern is confounding by indication, whereby the underlying maternal conditions prompting paracetamol use (eg, fever, infection, inflammation) may themselves be associated with paracetamol use.[13] Consequently, recent reviews[6,14] have specifically recommended that future research prioritize study designs that better account for unmeasured genetic and shared environmental confounding, such as sibling-controlled analyses.

The methodologic imperative for sibling-controlled design is reinforced by the strong familial aggregation of both ASD (heritability of approximately 80%) and attention-deficit/hyperactivity disorder (ADHD; heritability approximately 74%).[15,16] While conventional cohort studies have often reported positive associations on ASD and ADHD, the few existing sibling-controlled studies (n = 3) have consistently demonstrated a complete attenuation of these associations after controlling for shared familial factors.[12,17,18]

However, this critical sibling-controlled evidence remains sparse, derived exclusively from Nordic (Norwegian, Swedish) and East Asian (Japanese) cohorts.[12,17-19] While some of these studies used registry data not susceptible to recall bias, exposure assessment in the existing literature has been limited in granularity. The Swedish study evaluated dosage for a subset of the data,[12] but none have comprehensively assessed detailed exposure characteristics beyond a binary measure. Consequently, it remains unknown whether these null findings generalize to populations with different genetic, cultural, and clinical contexts.[20-22]

Therefore, our study aimed to address these gaps by leveraging a population-based mother-child cohort in Hong Kong. We used prospectively recorded electronic prescription records to enable detailed exposure assessment that is less susceptible to recall bias. This provided an opportunity for methodologic triangulation, whereby findings derived from objective prescription data in a new population can be compared with existing evidence based on self-report or other registry data—thereby enhancing robustness and generalizability. Using a sibling-controlled design to control for unmeasured familial confounding, we rigorously evaluated the associations between prenatal paracetamol exposure, including timing, pattern, and cumulative dose, on the risk of ASD and ADHD in offspring.

Key Points

Question Does prenatal exposure to paracetamol (acetaminophen) increase the risk of autism spectrum disorder (ASD) or attention-deficit/hyperactivity disorder (ADHD) in offspring after accounting for unmeasured familial confounding?

Findings From an initial cohort of 708 020 mother-child pairs in Hong Kong (approximately 43.3% with prenatal paracetamol exposure), a sibling-matched cohort was constructed of 124 333 children who were assessed for ASD and 97 285 for ADHD. Prenatal paracetamol exposure was not associated with an increased risk of either ASD or ADHD and these findings were consistent across paracetamol exposure timing, pattern, and dose.

Meaning These findings provide important reassurance regarding the safety of indicated paracetamol use during pregnancy.

Methods

The study protocol was approved by the institutional review board of the University of Hong Kong–Hospital Authority Hong Kong West Cluster (UW 20-051). Informed consent was waived because this study did not have patient contact and used anonymized data. This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Data Source

The Hong Kong Hospital Authority provides universal health coverage to a predominantly Chinese population of 7.5 million, overseeing all public health care services through a network of 43 hospitals and institutions, 49 specialist outpatient clinics, and 75 family medicine clinics (2025 data).[23] The clinical management system maintains comprehensive electronic health records, including demographic characteristics, diagnoses, prescriptions, laboratory results, and procedures, to provide real-time updates for routine clinical practice. Diagnoses and procedures were coded using the *International Classification of Diseases, Ninth Revision, Clinical Modification* (ICD-9-CM) and the *International Classification of Primary Care* (ICPC) codes. Prescription data are routinely verified by pharmacists, and previous validation studies have demonstrated high data accuracy.[24]

The robust mother-child linkage system, established for clinical management immediately and permanently after delivery, captures records for more than 500 000 mothers and

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660 000 infants (2023 data).[25] This database has been successfully used in numerous pharmacoepidemiologic studies, including investigations of neurodevelopmental disorders in relation to maternal drug exposures.[26-29]

Study Population and Design

We established an initial cohort of mother-child pairs with live births between January 1, 2001, and December 31, 2023, for mothers aged 15 to 50 years who were current Hong Kong residents. A valid linkage required an exact match of mother and child identification numbers and delivery date. We excluded pregnancies with missing linkage or incomplete key information (eg, infant sex, last menstrual period [LMP]), followed by miscarriages, perinatal death, and multifetal pregnancies, creating a singleton live birth cohort.

To strengthen association, we used a triangulation of approaches.[30] The primary analysis used a sibling-matched design. From the singleton live birth cohort, we identified all maternal siblings (ie, children sharing the same biological mother) and constructed a cohort of families with discordant prenatal paracetamol exposure (ie, at least 1 sibling exposed and another unexposed) (eFigure 1 in Supplement 1).[31] We also conducted a conventional cohort analysis and negative control exposure analysis (prepregnancy and postpregnancy) in the full population.[32,33]

Prenatal Paracetamol Exposure

The pregnancy period was defined as from the LMP to delivery, with LMP derived by subtracting clinically determined gestational age (based on first-trimester ultrasonography imaging) from the delivery date. Trimesters were defined as first (0-12 weeks), second (13-27 weeks), and third (28 weeks to delivery).

Prenatal paracetamol exposure was ascertained from prescription records in the public health care system; over-the-counter (OTC) purchases and private prescriptions were not captured. Prescription records provided detailed, preprescription information, including drug name, formulation, strength (mg), prescribed number of tablets per administration, administration frequency (times/d), and start and end dates. Paracetamol was identified using the British National Formulary (BNF), section 4.7.1. Data completeness for key fields was high (99.98%), enabling robust derivation of exposure timing, pattern, and dosage.

The primary exposure was any paracetamol use during pregnancy. To assess critical developmental windows (timing) and exposure patterns,[34] we created mutually exclusive categories: unexposed, sporadic (use in 1 trimester only), intermittent (use in any 2 trimesters), and persistent (use in all 3 trimesters). To assess for a biological gradient, we calculated mean daily dose (mg/d) as strength × dosage per administration × frequency, then categorized use as: none; low (<27.5 mg/d; <25th percentile); medium (27.5 to <75.2 mg/d; 25th to 75th percentile); and high (≥75.3 mg/d; >75th percentile).

Outcomes and Follow-Up

Primary outcomes were clinically diagnosed ASD, per *ICD-9-CM* code 299; or ADHD per code 314 or a filled prescription for an ADHD-specific medication licensed in Hong Kong (ie, methylphenidate, atomoxetine, or lisdexamfetamine; BNF, section 4.4.[26,35] Children were included in the ASD analyses only if they had 2 or more years of follow-up, and in ADHD analyses only if they had 5 or more years of follow-up. Follow-up began at delivery and continued until the first occurrence of outcome, death, or December 31, 2023 (eFigure 2 in Supplement 1).

Covariates

We adjusted for a comprehensive set of covariates selected a priori based on literature and clinical expertise including maternal characteristics at delivery, medical conditions and medication use before or during pregnancy, and child characteristics at birth (eTable 1 in Supplement 1).

Statistical Analysis

Sibling-Matched Analysis

We used stratified Cox proportional hazards regression (stratified by family) with cluster-robust standard errors to estimate hazard ratios (HRs) for the association between prenatal paracetamol exposure and ASD or ADHD. A 2-sided $P < .05$ was considered statistically significant. Data analysis was conducted from October 2, 2025, to March 24, 2026.

Subgroup Analysis Using the exposure metrics defined previously, we estimated risks by exposure patterns (sporadic, intermittent, and persistent), dose-response (categorized mean daily dose). Given the established male predominance in neurodevelopmental disorders (approximately 3:1 for ASD and 2:1 for ADHD),[36,37] we stratified by child sex. To account for temporal changes in diagnostic criteria and prescribing practices, we stratified by birth era (2001-2010 vs 2011 onwards). We also evaluated maternal age (≥35 years) as a potential modifier.

Sensitivity Analysis We evaluated the robustness of our findings to alternative exposure definitions: (1) sustained use (≥1 prescription of ≥7 consecutive days); and (2) cumulative use (total supply ≥7 days for all prescriptions).[38] Then, to mitigate potential confounding by indication, we excluded from the unexposed group those mothers who were prescribed alternative analgesics (oral opioids or nonsteroidal anti-inflammatory drugs [NSAIDs]) during pregnancy.[39]

Full Cohort Analysis

As part of our triangulation strategy, we reestimated the association in the full cohort using propensity score (PS) matching (1:1, calliper width = 0.05), and inverse probability of treatment weighting (IPTW), achieving covariate balance (standardized mean differences [SMD] < 0.10).[40,41] In addition, we conducted negative control exposure analyses using maternal paracetamol use in the year preceding pregnancy (prepregnancy) and in the year after delivery (postpregnancy). A nonnull association in these etiologically implausible windows would suggest that the positive associations observed in the full-cohort analysis are substantially biased by unmeasured or unknown confounding (eAppendix 1 in Supplement 1).[32,33]

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Bias Assessment

To address carryover effect, where an ASD or ADHD diagnosis in the first child influences paracetamol use in a subsequent pregnancy,[42] we fitted a logistic regression model adjusting for the first child's characteristics and family confounders (eAppendix 2 in Supplement 1). Finally, we performed a quantitative bias analysis to assess the extent to which exposure measurement error[43]—arising from uncaptured OTC use, private prescriptions, or variation in patient adherence to prescribed regimens—may have attenuated the observed associations toward the null in the sibling-matched and full-cohort analyses (eAppendix 3 in Supplement 1).

Results

Of an initial 708 020 mother-child pairs, 306 955 (approximately 43.3%) had prenatal paracetamol exposure. From these, we constructed 2 outcome-specific sibling-matched cohorts. For ASD analysis, the cohort comprised 124 333 children (mean [SD] age, 9.3 [4.7] years; 61 775 females [49.7%] and 62 558 males [50.3%]) with discordant prenatal paracetamol exposure. Median (IQR) follow-up was 10.2 (6.8-14.1) years, during which 3445 children were diagnosed with ASD (2.8%). For ADHD analysis, the cohort comprised 97 285 children (mean [SD] age, 7.6 [4.2] years; 48 455 females [49.8%] and 48 830 males [50.2%]) with discordant prenatal paracetamol exposure. Median (IQR) follow-up was 11.3 (8.3-14.9) years, during which 5168 children were diagnosed with ADHD (5.3%) (**Figure 1**).

Within the same mother, pregnancies with prenatal paracetamol dispensing were associated with older maternal age at delivery, higher prevalence of preexisting conditions (eg, psychiatric disorders), and more common indications for analgesic use (eg, infection, fever, chronic pain), compared with pregnancies without paracetamol dispensing. Pregnancies with paracetamol dispensing were also more likely to involve concurrent dispensing of other analgesics (aspirin, NSAIDs, opioids) (**Table 1**).

Prenatal Paracetamol Exposure and ASD or ADHD in Offspring

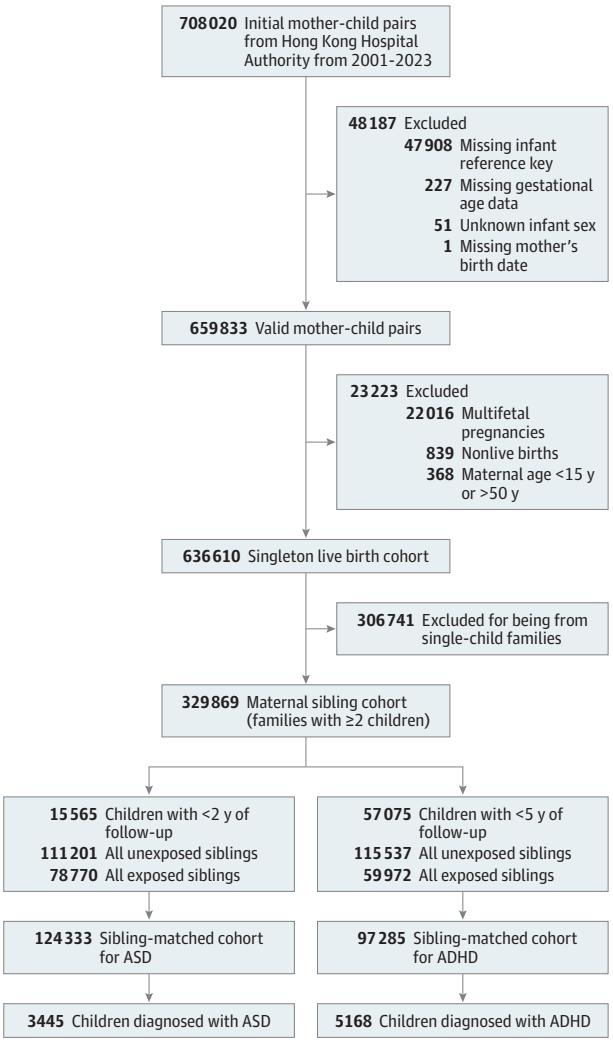
Primary and Subgroup Analyses (Sibling-Matched Cohort)

In the primary sibling-matched analysis, prenatal paracetamol exposure was not associated with ASD (adjusted HR, 1.00; 95% CI, 0.91-1.11) or ADHD (adjusted HR, 1.01; 95% CI, 0.93 to 1.08). There was no evidence of a critical exposure window by trimester nor meaningful variation in risk by exposure patterns (sporadic, intermittent, persistent) or categorized mean daily dose, with no dose-response relationship for either outcome (**Figure 2** and **Figure 3**). Subgroup analyses revealed no evidence of effect modification by child sex, birth era (2001-2010 vs 2011 onwards), or maternal age (≥35 years) (eTable 2 in Supplement 1).

Sensitivity Analyses (Sibling-Matched Cohort)

All sensitivity analyses within the sibling-matched framework yielded null associations. This included when exposure was redefined based on sustained use (≥7 consecutive days) or cumulative use (total supply ≥7 days), and when the unexposed comparator was restricted to women not dispensed alternative analgesics (opioids or NSAIDs) (eTable 3 in Supplement 1).

Figure 1. Flow Diagram of Study Population Selection for Sibling-Matched Analysis of Prenatal Paracetamol Exposure and Autism Spectrum Disorder (ASD) and Attention-Deficit/Hyperactivity Disorder (ADHD) in Offspring



Sibling-matched cohort formed using data from the Hong Kong Hospital Authority (2001-2023) database. Using a discordant exposure design, exposed and unexposed siblings within the same family were matched to control for unmeasured familial factors. Prenatal paracetamol exposure was defined as any maternal use recorded between the last menstrual period and delivery.

Full Cohort Comparison

In contrast to the sibling-matched results, conventional Cox regression in the full cohort (singleton live births) showed positive association between prenatal paracetamol exposure and ASD (HR, 1.17; 95% CI, 1.13-1.20) and ADHD (HR, 1.23; 95% CI, 1.20-1.27) (**Table 2**). Negative control analyses of prepregnancy paracetamol exposure also yielded positive associations

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Table 1. Characteristics of Mothers and Children in the Sibling-Matched Cohort Study of Prenatal Paracetamol Exposure and Autism Spectrum Disorder (ASD) or Attention-Deficit/Hyperactivity Disorder (ADHD) Risk

Characteristic	No. (%)			
	ASD		ADHD	
	Dispensed paracetamol (n = 62 041)	Not dispensed paracetamol (n = 62 292)	Dispensed paracetamol (n = 48 215)	Not dispensed paracetamol (n = 49 070)
Mother's characteristics at delivery				
Age, mean (SD), y	31.4 (5.1)	29.6 (5.0)	31.0 (5.1)	29.7 (5.0)
Year of delivery (SD)	2014 (5)	2012 (5)	2012 (4)	2011 (4)
Parity				
0 (Nulliparous)	16 152 (26)	33 071 (53)	14 099 (29)	24 435 (50)
>0 (Multiparous)	45 889 (74)	29 221 (47)	34 116 (71)	24 635 (50)
Delivery mode				
Cesarean	12 816 (21)	12 399 (20)	9590 (20)	9970 (20)
Vaginal	49 117 (79)	49 805 (80)	38 549 (80)	39 029 (80)
Other	108 (0.2)	88 (0.1)	76 (0.2)	71 (0.1)
Conditions (1 y before or during pregnancy)				
ASD	3 (<0.1)	2 (<0.1)	2 (<0.1)	2 (<0.1)
ADHD	37 (<0.1)	35 (<0.1)	26 (<0.1)	28 (<0.1)
Intellectual disability	69 (0.1)	66 (0.1)	56 (0.1)	55 (0.1)
Psychiatric disorders	4469 (7.2)	3420 (5.5)	3307 (6.9)	2739 (5.6)
Infection	36 687 (59)	32 787 (53)	28 735 (60)	25 923 (53)
Fever	16 (<0.1)	7 (<0.1)	12 (<0.1)	7 (<0.1)
Chronic pain	1158 (1.9)	567 (0.9)	991 (2.1)	459 (0.9)
Headache	431 (0.7)	164 (0.3)	359 (0.7)	132 (0.3)
Migraine	53 (<0.1)	30 (<0.1)	41 (<0.1)	26 (<0.1)
Asthma	392 (0.6)	282 (0.5)	316 (0.7)	231 (0.5)
Rheumatoid arthritis	12 (<0.1)	13 (<0.1)	8 (<0.1)	9 (<0.1)
Dyslipidemia	14 (<0.1)	6 (<0.1)	6 (<0.1)	3 (<0.1)
Chronic kidney disease	353 (0.6)	221 (0.4)	280 (0.6)	181 (0.4)
Hypertensive disease	146 (0.2)	88 (0.1)	106 (0.2)	76 (0.2)
Gestational hypertension	1702 (2.7)	1475 (2.4)	1280 (2.7)	1195 (2.4)
Diabetes (type 1 or 2)	1020 (1.6)	748 (1.2)	772 (1.6)	581 (1.2)
Gestational diabetes	6400 (10)	4802 (7.7)	4776 (9.9)	3744 (7.6)
Medications (dispensed within 1 y before or during pregnancy)				
Aspirin	1088 (1.8)	436 (0.7)	592 (1.2)	264 (0.5)
NSAIDs	6251 (10)	1607 (2.6)	4091 (8.5)	1134 (2.3)
Opioid analgesics	12 544 (20)	4019 (6.5)	8391 (17)	2853 (5.8)
Antidepressants drugs	882 (1.4)	493 (0.8)	676 (1.4)	396 (0.8)
Antimigraine drugs	6 (<0.1)	6 (<0.1)	5 (<0.1)	6 (<0.1)
Antiepileptic drugs	138 (0.2)	104 (0.2)	103 (0.2)	87 (0.2)
Lipid-regulating drugs	6 (<0.1)	0 (0)	4 (<0.1)	0 (0)
Antihypertensives	1490 (2.4)	1026 (1.6)	1113 (2.3)	801 (1.6)
Glucose-lowering agents	670 (1.1)	392 (0.6)	485 (1.0)	297 (0.6)
Child's characteristics				
Sex				
Female	30 500 (49)	31 275 (50)	23 845 (49)	24 610 (50)
Male	31 541 (51)	31 017 (50)	24 370 (51)	24 460 (50)
Birth trauma	166 (0.3)	200 (0.3)	147 (0.3)	164 (0.3)
Preterm birth (<37 wk's gestation)	3742 (6.0)	3990 (6.4)	2844 (5.9)	3255 (6.6)
Birth weight, mean (SD) g	3153 (460)	3137 (465)	3155 (459)	3141 (471)
Apgar score at 1 min, mean (SD)	8.68 (0.97)	8.57 (0.96)	8.73 (0.98)	8.59 (0.98)
Apgar score at 5 min, mean (SD)	9.46 (0.73)	9.43 (0.72)	9.51 (0.70)	9.45 (0.71)
Conditions diagnosed in follow-up period in childhood				
Cases	1629 (2.6)	1816 (2.9)	2385 (4.9)	2783 (5.7)
Years of follow-up, median (IQR)	9.17 (5.49-13.06)	11.06 (7.91-14.94)	10.68 (7.74-14.02)	11.98 (8.96-15.58)

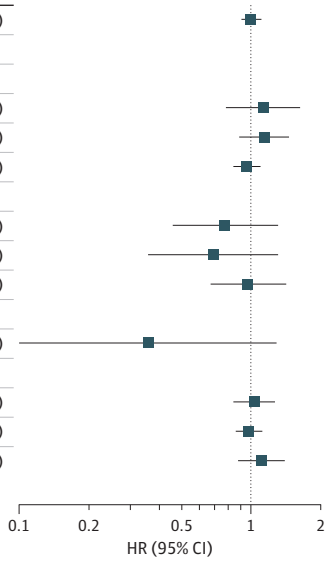
Abbreviation: NSAIDs, nonsteroidal anti-inflammatory drug.

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Figure 2. Dot Plot of Sibling-Matched Analysis for Prenatal Paracetamol Exposure and Autism Spectrum Disorder (ASD) Risk in Offspring

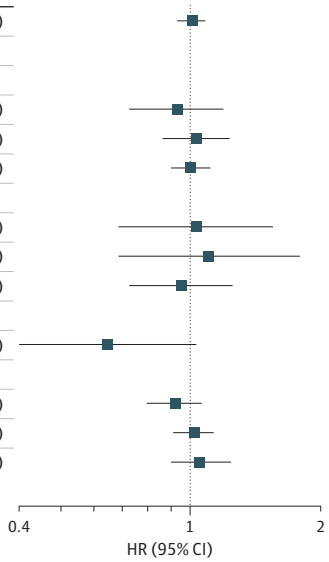
Group	Children, No.		HR (95% CI)	*P* value
	Exposed	Not exposed		
Primary analysis	62041	62292	1.00 (0.91-1.11)	.99
By exposure timing and pattern				
Sporadic (1 trimester only)				
Trimester 1 only	4179	4707	1.13 (0.78-1.63)	.51
Trimester 2 only	7110	7941	1.14 (0.89-1.46)	.30
Trimester 3 only	41548	43064	0.96 (0.84-1.10)	.57
Intermittent (any 2 trimesters)				
Trimesters 1 and 2	1518	1735	0.77 (0.46-1.31)	.34
Trimesters 1 and 3	2424	2150	0.69 (0.36-1.31)	.25
Trimesters 2 and 3	4245	4682	0.97 (0.67-1.42)	.89
Persistent (all 3 trimesters)				
All trimesters	1291	1423	0.36 (0.10-1.29)	.12
By dose-response (mean daily dose)				
Low	17101	18339	1.04 (0.84-1.27)	.74
Medium	32164	34129	0.98 (0.86-1.12)	.78
High	12756	13519	1.11 (0.88-1.40)	.38



Hazard ratios (HRs) and 95% CIs are shown for the associations between prenatal paracetamol exposure and ASD in offspring, stratified by exposure timing, pattern, and cumulative dose (categorized as low [<27.5 mg/d; <25th percentile], medium [<75.2 mg/d; 25th-75th percentile], and high [≥75.3 mg/d; >75th percentile]). The vertical dotted line indicates HR of 1.0 (no effect).

Figure 3. Dot Plot of Sibling-Matched Analysis for Prenatal Paracetamol Exposure and Offspring Attention-Deficit/Hyperactivity Disorder (ADHD) Risk

Group	Children, No.		HR (95% CI)	*P* value
	Exposed	Not exposed		
Primary analysis	48215	49070	1.01 (0.93-1.08)	.88
By exposure timing and pattern				
Sporadic (1 trimester only)				
Trimester 1 only	3869	4349	0.93 (0.72-1.19)	.55
Trimester 2 only	6549	7279	1.03 (0.86-1.23)	.78
Trimester 3 only	30520	31841	1.00 (0.90-1.11)	.86
Intermittent (any 2 trimesters)				
Trimesters 1 and 2	1407	1598	1.03 (0.68-1.55)	.89
Trimesters 1 and 3	1608	1805	1.10 (0.68-1.79)	.69
Trimesters 2 and 3	3259	3586	0.95 (0.72-1.25)	.70
Persistent (all 3 trimesters)				
All trimesters	1003	1104	0.64 (0.40-1.03)	.07
By dose-response (mean daily dose)				
Low	14317	15418	0.92 (0.79-1.06)	.24
Medium	23850	25392	1.02 (0.91-1.13)	.76
High	10037	10649	1.05 (0.90-1.24)	.52



Hazard ratios (HRs) and 95% CIs are shown for the associations between prenatal paracetamol exposure and ADHD in offspring, stratified by exposure timing, pattern, and cumulative dose (categorized as low [<27.5 mg/d; <25th percentile], medium [<75.2 mg/d; 25th-75th percentile], and high [≥75.3 mg/d; >75th percentile]). The vertical dotted line indicates HR of 1.0 (no effect).

for ASD (HR, 1.12; 95% CI, 1.08-1.17) and ADHD (HR, 1.24; 95% CI, 1.20-1.28). Similarly, positive associations were observed for postpregnancy exposure (eTables 4 and 5 in Supplement 1). These positive associations were consistent across PS-matched and IPTW-adjusted models (Table 2; eTables 6 and 7 in Supplement 1).

Bias Assessment

Logistic regression assessing carryover effects (where a prior child's diagnosis influences subsequent exposure) suggested this bias was probably absent for both ASD and ADHD (eTables 8 and 9 in Supplement 1). Quantitative bias analysis demonstrated that nondifferential exposure measurement error attenuated effect estimates in sibling-matched and conventional full-cohort analyses, with slightly greater attenuation in the sibling-matched design. For effect sizes within the range observed in our main analysis, the degree of attenuation was minimal, suggesting that nondifferential measurement error in paracetamol exposure was unlikely to have masked a strong true association (eTable 10 in Supplement 1).

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Table 2. Association of Prenatal Paracetamol Exposure and Risk of Autism Spectrum Disorder (ASD) and Attention-Deficit/Hyperactivity Disorder (ADHD) in Offspring: Comparison of Sibling-Matched, Conventional Cohort, and Negative Control Analyses

Study design and statistical model	ASD			ADHD		
	No. of exposure	No. of control	HR (95% CI)	No. of exposure	No. of control	HR (95% CI)
Primary analysis: sibling-matched						
Cox regression	62 041	62 292	1.00 (0.91-1.11)	48 215	49 070	1.01 (0.93-1.08)
Conventional cohort						
Cox regression	259 963	337 974	1.17 (1.13-1.20)	196 002	320 926	1.23 (1.20-1.27)
Propensity-score matching	185 894	185 894	1.15 (1.11-1.20)	163 732	163 732	1.23 (1.19-1.27)
IPTW	242 021	333 323	1.16 (1.11-1.20)	196 002	320 926	1.23 (1.20-1.27)
Negative control: prepregnancy exposure						
Cox regression	108 862	489 075	1.12 (1.08-1.17)	90 195	426 733	1.24 (1.20-1.28)
Propensity-score matching	107 023	107 023	1.21 (1.15-1.27)	88 903	88 903	1.27 (1.22-1.33)
IPTW	108 549	488 152	1.23 (1.18-1.29)	89 929	425 942	1.33 (1.29-1.38)

Abbreviations: HR, hazard ratio; IPTW, inverse probability of treatment weighting.

Negative control exposure analysis refers to maternal paracetamol use in the year preceding pregnancy. All models were adjusted for maternal sociodemographic (age, year, parity, delivery mode), maternal diagnoses and comedications during pregnancy, and neonatal characteristics (sex, birth weight, Apgar scores, birth trauma, preterm birth). A full list of covariates is provided in the eTable 1 in Supplement 1.

Discussion

This study triangulated evidence using a sibling-matched design, a conventional cohort analysis, and a negative control exposure analysis in the full cohort. Our primary sibling-matched analysis, which accounted for unmeasured familial confounding, found no association between prenatal paracetamol exposure and the risk of ASD or ADHD in offspring. This null finding was robust across exposure timing, patterns, and doses. In contrast, the conventional cohort analysis, despite extensive adjustment for measured confounders, replicated the positive associations frequently reported in prior literature.[6] The persistence of positive associations after PS and IPTW adjustment suggests that unmeasured familial factors, particularly genetic predisposition and health literacy, are likely the primary source of residual confounding. These factors are inherently accounted for in the sibling design but cannot be fully captured by conventional administrative data. Most critically, the negative control analyses also yielded positive associations with both prepregnancy and postpregnancy exposure—periods when an associated effect of paracetamol on fetal neurodevelopment is biologically implausible. Collectively, these findings suggest that the positive signal observed in both conventional and negative control analyses reflect residual familial confounding, rather than a true pharmacologic effect of prenatal paracetamol exposure.[44,45]

Together, these study findings—positive associations in conventional cohort analyses but null results in sibling-matched analyses—reconcile the conflicting literature. The consistency of null findings from robust sibling-matched designs across different populations,[11-13] now provides strong evidence against a causal link. The collective evidence supports current regulatory guidance on the safety of indicated antenatal paracetamol use. Paracetamol remains a safe and essential analgesic and antipyretic during pregnancy, whereas alternatives such as NSAIDs and opioids carry well-documented risks, including teratogenicity, oligohydramnios, neonatal abstinence syndrome, and kidney damage.[46-48] Unwarranted reluctance to use paracetamol could lead to undertreatment of pain and fever, or the use of more harmful alternatives, both posing risks to the pregnancy and developing fetus.[49] Clinicians should reassure patients that paracetamol is safe when used as clinically indicated, at the lowest effective dose for the shortest necessary duration.

Strengths and Limitations

A key strength of our study is the application of complementary analytical designs to address pervasive unmeasured confounding, including confounding by indication and shared familial factors. Given the strong familial aggregation of ASD and ADHD,[15,16] our sibling-matched design inherently controlled for stable genetic and environmental factors shared within families. Our null results align with sibling-matched studies from Norway, Sweden, and Japan, which also found attenuated associations after controlling for familial factors.[6,12,17,18] This convergence across diverse populations strengthens the evidence against a causal effect.

Moreover, this study extends the existing literature in several ways. First, extended follow-up (up to 23 years) provides stable estimates with sufficient statistical power, as reflected in the narrow confidence intervals of our null findings. Second, using prospectively recorded electronic prescriptions records, we moved beyond binary ever-use assessment that characterized most prior sibling-matched studies.[12,17] This objective data source enabled granular analyses of exposure timing, patterns of use, and dose-response relationships.[12,18] Third, these detailed analyses offer timely evidence to address specific clinical concerns regarding paracetamol used during pregnancy.

Several limitations warrant consideration. First, while the sibling-matched design controls for shared familial confounders, it has inherent limitations.[31] Although we adjusted for measured nonshared factors (eg, age, birth order), residual nonshared confounding may persist and could even be amplified in sibling comparisons.[50] The design generally assumes an

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absence of carryover effects between siblings[31,42] and our sensitivity analyses suggested that this assumption holds true, although it cannot be definitively ruled out. Measurement error can also be amplified in sibling comparisons, as exposure misclassification may have greater impact in within-family than between-family analyses.[50] Finally, restriction to families with exposure-discordant siblings could limit generalizability to the general population.

Second, our data capture prescription paracetamol use from the public health care system only. Paracetamol is also widely available OTC that also may be prescribed in private settings. Uncertainty in the timing of medication use relative to prescription dates means that some prescriptions may not reflect actual use during pregnancy. This exposure misclassification would be nondifferential and therefore would bias results toward the null. Additionally, prescription users may differ from OTC users in their underlying indications and symptom severity. Prescription use likely reflects more clinically meaningful conditions that led to an outpatient physician visit, whereas OTC use captures sporadic, self-managed symptoms. Notably, in Hong Kong, it is common for physicians to prescribe OTC medications that are then dispensed in clinic-attached pharmacies. Therefore, the extent to which our findings apply to OTC use remains uncertain. However, 2 factors mitigate this concern. First, our quantitative bias analysis indicated that such misclassification is unlikely to fully explain the null findings. Second, within a sibling-matched design, maternal behavior regarding uncaptured medication use is likely consistent across pregnancies, reducing bias in within-family comparisons. Nevertheless, we acknowledge that some degree of exposure misclassification is inevitable given the lack of OTC purchase data, and the possibility of residual bias toward the null cannot be entirely ruled out.

Conclusions

This population-based cohort study adds evidence from a large Chinese population to the growing international consensus that prenatal paracetamol exposure is not a major risk factor for ASD or ADHD in offspring. While this study directly addresses recent public concerns regarding these specific outcomes, future research should investigate other neurodevelopmental outcomes (eg, intellectual disability). More broadly, future studies of prenatal medication safety should, when possible, use sibling-matched designs and methodologic triangulation to address the persistent challenge of unmeasured familial confounding.

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Invited Commentary

WOMEN'S HEALTH

Acetaminophen Use in Pregnancy and Neurodevelopmental Outcomes—Reassuring Evidence From a Matched-Sibling Study

Brian K. Lee, PhD; Viktor H. Ahlqvist, PhD

In September 2025, the US administration held a press conference stating that its ongoing investigation into the causes of autism had identified maternal use of acetaminophen (paracetamol) during pregnancy as a potential causal factor. This was concerning given that acetaminophen is among the most used medications in pregnancy for the management of pain and fever, in part because alternative options are limited or contraindicated (eg, nonsteroidal anti-inflammatory medications and opioids).

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Assessing pregnancy medication safety with regard to long-term neurodevelopmental outcomes is challenging. This research requires large sample sizes, long-term follow-up into childhood, and is highly susceptible to confounding. The causes of neurodevelopmental conditions are multifactorial and often strongly correlated with maternal health, behaviors, and genetic predisposition.[1] Addressing these sources of bias is necessary before any inference that acetaminophen causes neurodevelopmental outcomes can be made. One approach to attempt to eliminate this confounding is the sibling comparison design, in which exposed and unexposed siblings are compared directly. By leveraging within-family comparisons, this design accounts for all confounding factors shared by siblings, including maternal genetics and stable environmental influences.[2]

In the context of acetaminophen and neurodevelopmental outcomes, 4 prior studies, conducted in Japan, Norway, Sweden, and Taiwan, have applied sibling comparison designs and consistently found no increased risk when comparing siblings.[3,4] However, the Japanese and Norwegian studies were relatively small, and although the Swedish and Taiwanese studies were larger, they warrant replication in independent populations before definitive conclusions can be drawn.[3-5] In this issue of *JAMA Internal Medicine*, Luo et al[6] extend the literature by reporting findings from a large-scale study conducted in Hong Kong.

Evidence Against a Causal Association

Using comprehensive health records from the Hong Kong Hospital Authority and a mother-child linkage, Luo et al[6] identified 597 937 and 516 928 liveborn singleton children for analyses of autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD), respectively, after excluding those with insufficient follow-up or missing covariate data. Approximately 43.3% of these pregnancies involved dispensed acetaminophen.

In analyses of the full population cohort, adjusting for measured confounders, prenatal acetaminophen exposure was associated with increased risks of autism (hazard ratio [HR], 1.17; 95% CI, 1.13-1.20) and ADHD (HR, 1.23; 95% CI, 1.20-1.27), findings consistent with those of prior studies. Interestingly, maternal acetaminophen use in the year prior to pregnancy was associated with a similar increase in risk, suggesting that these associations reflect residual confounding rather than a causal effect, given that prepregnancy exposure would not be expected to affect offspring neurodevelopment (ie, a negative control exposure).

Within this cohort, the authors further identified 124 333 and 97 285 exposure-discordant sibling pairs for analyses of ASD and ADHD, respectively. In sibling comparison analyses, the associations attenuated to the null (ASD: HR, 1.00 [95% CI, 0.91-1.11]; ADHD: HR, 1.01 [95% CI, 0.93-1.08]), consistent with confounding observed in the negative control analysis.

Taken together, the modest associations observed in the full cohort, the comparable estimates for prepregnancy exposure (negative control), and the attenuation to the null in sibling comparisons provide converging evidence against an associated causal relationship between prenatal acetaminophen exposure and ASD or ADHD.

Strengths and Limitations of the Study

A key advance over prior work is the detailed characterization of acetaminophen use during pregnancy. Whereas earlier studies typically have treated acetaminophen use as a single exposure across pregnancy, Luo et al[6] assessed multiple patterns of use, including trimester-specific use (restricted to a single trimester), intermittent use (across 2 trimesters), and

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